

Chapter 5

A Brief Review of the Cleaning Process for Electronic Device Fabrication

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1 INTRODUCTION

In the fabrication of electronic devices, undesired organic or inorganic contaminant materials must be removed; otherwise, they will have a serious effect on the efficiency of the device, as well as its functionality. The purpose of the cleaning process is to eliminate contaminants that can affect the performance characteristic of the electronic devices.

The concept of cleaning is applied to any process that can remove traces of contamination in the form of organic and inorganic particles, traces of metals after the etch process, traces of photoresist after mask cleaning, and traces of slurry residue after a chemical mechanical planarization (CMP) process, and lastly, removal of by-products of cleaning processes after dry/wet cleaning.

Therefore, different cleaning processes with their advantages, as well as their disadvantages, after metal, silica, and silicon etch, must be considered before selecting a cleaning procedure. Another important issue in cleaning is selection of the procedure to find out how the cleaning solutions need to be exposed on the surface of wafers and disks during removal of contamination. These procedures may be considered as use of a stationary bath versus a stream bath, or use of spray tools during exposure of the cleaning solution on the surface of wafers and disks. The details of some important cleaning processes are explained in the following sections.

2 CLEANING OF INORGANIC CONTAMINATION

Advances in current electronic manufacturing technology have made the removal of contaminating particles, with sizes as small as nanometers, crucial in making high yield electronic components. The existence of unwanted particles during electronic chip manufacturing not only would contribute to short circuits, but they would also cause corrosion, and hence, would reduce the device's lifetime.

In general, contamination may be divided into two major groups: organic and inorganic. Organic contaminations, such as particles or traces of thin films, can further be divided into polar or nonpolar, as well as a combination of both. Inorganic contaminations may also be divided into two major groups: polar (ionic) or nonpolar (nonionic).

To understand the cleaning process for each type of particle, they may be characterized into the following three categories:

- Neutral particles (no bonding with the surface).
- Ionically bonded particles (electrostatic bonding).
- Covalently bonded particles (molecular bonding).

2.1 Neutral Particles

Neutral particles do not have a strong interaction tendency, or bonding structure with the surface (although they do possess a type of van der Waals' interaction

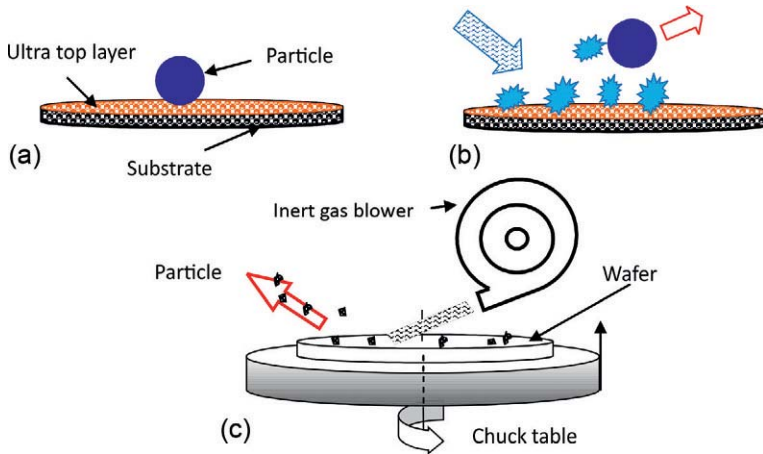


FIGURE 5.1 A schematic diagram showing removal of neutral charged particles. (a) and (b) DI-water rinsing; (c) application of pure argon gas to blow off the particles from the surface.

which is a weak interaction force), and thus, they can freely move to any location on the surface.

Their positions on the surface are neither permanent nor fixed. As such, they can easily be removed from the surface by a simple DI-water wash or by blowing the surface with an inert gas, such as argon (Fig. 5.1).

2.2 Ionically Bonded Particles

The second category of particles is the ionically bonded particles. These particles, in form of positive or negative charges, are bonded to the surface functional groups through electrostatic forces. To remove these particles from the surface, one may either use a repulsion force or apply a chelating agent that carries a functional group stronger than the surface functional group. The cleaning processes for ionic particles may be divided into the following:

- Ionic surface repulsion process (ISRP).
- Ionic trapping process (ITP).

2.2.1 Ionic SRP

In the ionic SRP technique, a cleaning solution with a high pH (excess hydroxide ions, negative ions) (Fig. 5.2) or a low pH (excess hydronium ions, positive ions) (Fig. 5.3), is applied to the surface.

As shown in Fig. 5.2, positively charged particles may become attracted to any surface that carries a negative charge (through electrostatic forces). When such particles are exposed to a cleaning solution with a high concentration of hydroxide ions (negative ions), they will be surrounded by hydroxide ions,

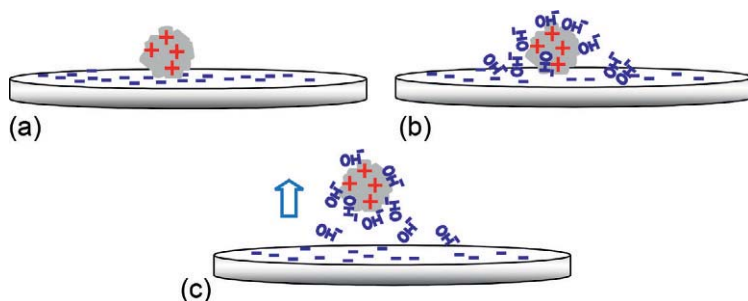


FIGURE 5.2 Immediate formation of similar charges on the surfaces of both substrate and particle to separate particles from the substrate surface by electro-repulsion force. (a) Shows position of one positive particle ion on the substrate before exposure to a cleaning solution. (b) Shows the instantaneous formation of similar charges on the surfaces of both substrate and particle after the surface has been exposed to the cleaning solution. (c) Shows separation of the particle from the surface due to electro-repulsion forces.

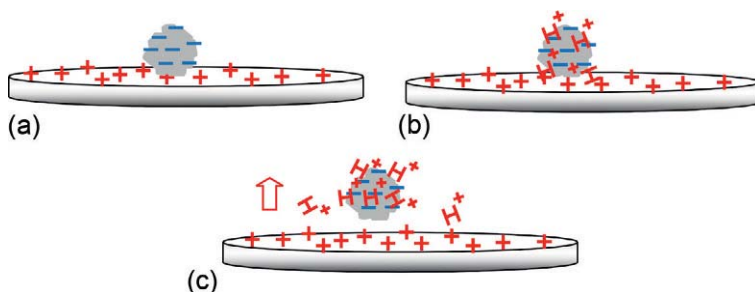


FIGURE 5.3 Effect of low pH on the substrate surface containing particles. (a) Position of one negative particle ion on the substrate before exposure to a cleaning solution. (b) Instantaneous formation of similar charges on the surface of both the substrate and the particle after the surface is exposed to the cleaning solution. (c) Separation of the particle from the surface due to positive electro-repulsion forces.

thereby reducing the charge density of these positive charges. In this process, due to the existence of several layers of hydroxide ions around each positive charge, the overall absolute charge around the positive particles will change from positive to negative, and as a result, the attraction force between the particles and the surface will change to repulsion forces. As such, the particles will no longer attach to the surface, and the surface will become particle free. In the same manner, particles with negative charges will be freed from the surface, as shown in Fig. 5.3.

As a general rule, there are always some restrictions during the cleaning process. When a cleaning solution is marked as having a high or low pH (high hydroxide ions or high hydronium ions), there is always the possibility that the cleaning solution will react with the surface and change the surface thickness as a result of surface etching. Therefore, wet clean personnel must pay

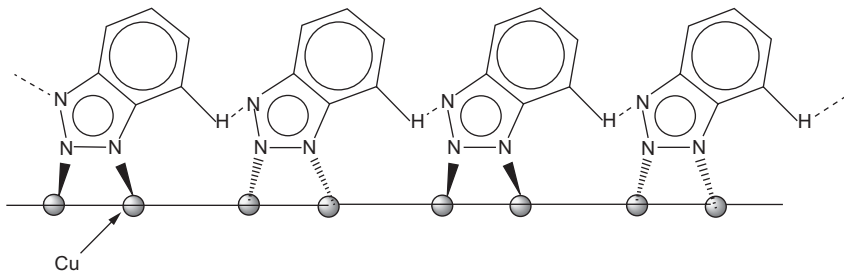


FIGURE 5.4 Structure of the first chemisorbed layer of 1,2,3-benzotriazole as corrosion inhibitors on a copper surface.

attention to the possible chemical reactions between the surface and the cleaning solutions. With today's electronic manufacturing technology, most devices are quite dense with thin electronic lines, thus, the cleaning process becomes quite complex. As a rule of thumb, the wet cleaning users must be aware of the chemical composition of the cleaning solution, as well as its pH, before committing to any cleaning solution candidate. Currently, chemical manufacturing companies are able to control surface etch (or surface corrosion) to a certain degree, by controlling the pH of the cleaning solution, or by using corrosion inhibitors (see Fig. 5.4).

Corrosion inhibitors are compounds that, when added to the cleaning solutions, protect the possibly easy-to-corrode surface from corrosion [1,2].

2.2.2 Ionic Trapping Process

In the ionic trapping process (ITP) the use of some chelating agent in the solution is another way to protect the surface for minimum etch and to create almost no change on the surface roughness by providing electrostatic bonding on the surface of the substrate (Fig. 5.5).

Depending on the chemical structure of the chelating agent for ITP, the chelating agent not only can shield the metal surface from further etching, but they can also trap some ions that are already preremoved by the cleaning solution through its dissolution or its zeta potential repulsion process. Therefore, by applying ITP cleaning, not only can one protect the surface from further etching,

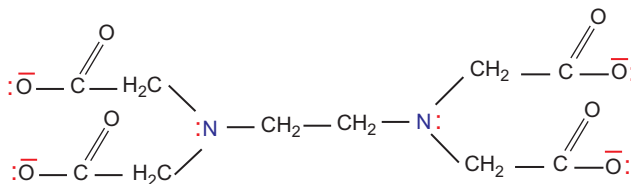


FIGURE 5.5 Molecular structure of ethylenediaminetetraacetic acid (EDTA) as a chelating agent.

but it can also eliminate the possibility of bonding between trace metal ions and the surface.

2.3 Covalently Bonded Particles

In case of strongly bonded particles (covalent bonds), the contaminating particles may be divided into two different groups as follows:

- Foreign bonded particles (FBPs).
- Native bonded particles (NBPs).

2.3.1 Foreign Bonded Particles

The FBPs are particles in which the molecular structure is different from the molecular structure of the surface. If the particles are FBP, the molecular structures must be known before the selection of any cleaning solution. To identify the molecular structure of these particles on the surface, one may use the EDX (energy dispersive X-ray spectroscopy) technique in order to determine the nature of these particles. Based on the EDX information, one should select a cleaning solution that removes these particles with minimum damage to the substrate surface. For cleaning FBP contamination, the best process is to find a cleaning solution that removes these particles without a chemical reaction with the surface (Fig. 5.6).

During the selection of a cleaning solution for the FBP contamination, it is very important to pay close attention to the chemical by-products of the cleaning process. Usually, the chemical by-products of dissolution are new ionic species that should be removed immediately. If these newly created by-products are not removed from the surface, they can cause short circuits or corrosion spots on a device. For example, when an acidic solution is utilized to remove

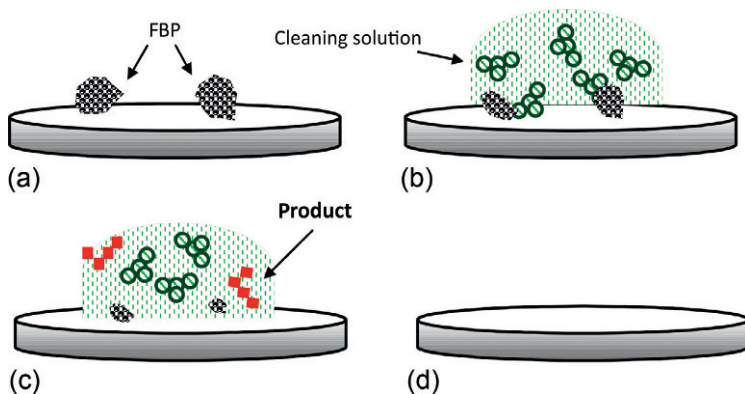


FIGURE 5.6 Removing FBP contamination from the surface by a chemical dissolution process. (a) FBP contamination on the substrate. (b) Exposure of FBP contamination to the cleaning solution. (c) Dissolution of FBP contamination by the cleaning solution. (d) Substrate after cleaning.

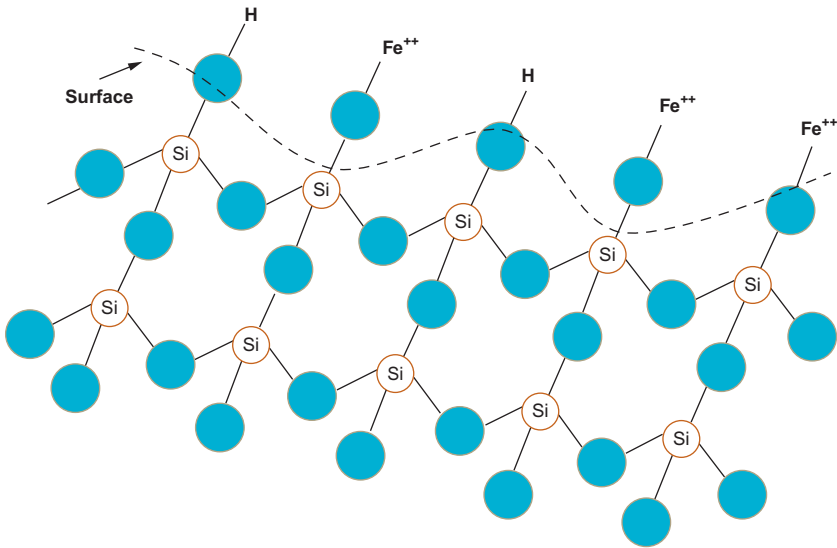
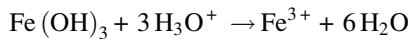


FIGURE 5.7 Rebinding the by-product of the cleaning solution (iron ions) on the surface.

iron hydroxide particles from the surface by a dissolution process, its by-product will be iron ion (Fe^{3+}).



These ions, if not removed from the surface, may become bonded to the surface in another manner, as shown in Fig. 5.7.

In case one is unable to find a good cleaning solution for dissolving foreign particles, operators may choose an undercut process, as shown in Fig. 5.8. In the undercut process, a solution that reacts with the surface, but not with the particles themselves, must be used. In this process, the surface underneath the particles will be removed to some degree in order to release attached particles from the surface. Subsequently, these particles can be removed from the surface through the zeta potential repulsion mechanism.

Thus, the chemical composition of the cleaning solutions must be selected in such a way not only to dissolve the FBP contamination but also to create a strong zeta potential on the surface in order to repel iron ions (the above example) away from the surface.

2.3.2 Native Bonded Particles

With native bonded particles (NBP) contamination, where the molecular structure of the particle is the same as the molecular structure of the surface, the cleaning process is completely different. To clean NBP from the surface, it is necessary to use a two-step cleaning process: by applying a CMP process with appropriate

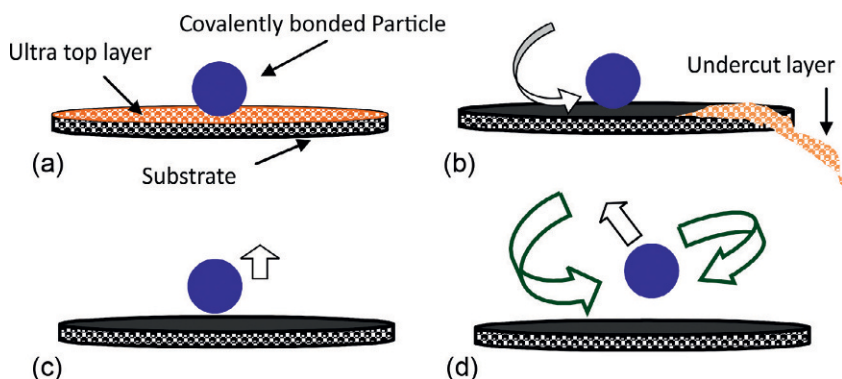


FIGURE 5.8 Removing NBP contamination from the surface by the undercut process. (a) NBP contamination on the substrate surface. (b) Effect of the undercut process on the top layer of the substrate surface by the cleaning solution. (c) Particle lift-off by elimination of surface bonded particle. (d) Movement of particle far away from the substrate surface by the electro-repulsive force created by the cleaning solution.

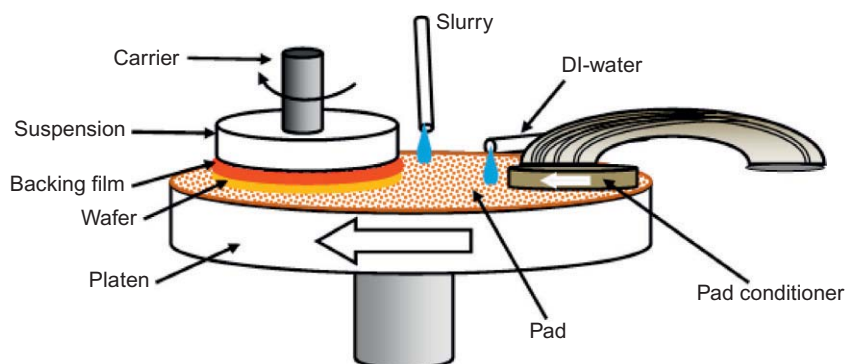
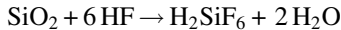


FIGURE 5.9 Simple CMP process for removing NBP contamination by surface polishing system.

slurry in order to remove these particles, and subsequently by utilizing another cleaning solution in order to remove the residues from the CMP process (Fig. 5.9).

Therefore, with NBP contamination, when the particle structure is the same, or close to the molecular structure of the surface, the cleaning process is more complex and sometimes requires several different chemistries in order to remove all CMP residues. The CMP process by itself is a great technique to reduce the surface roughness, but unfortunately this technique also generates a significant amount of particle contamination with different sizes on the surface of the polished substrates. Usually, most of the leftover particles on the surface, after a CMP process, are slurry particles in the form of either ions or molecules. Therefore, a post-CMP cleaning process is needed in order to

remove these slurry residues from the surface. The most common slurries are a mixture of aluminum oxide (Al_2O_3) with colloidal silica (SiO_2). A cleaning solution with negative zeta potential, with a pH of 12-13, is most commonly recommended to remove the majority of these slurry residues followed by exposure to a 0.1-5% HF solution. The selection of HF concentration as a cleaning solution depends on the concentration of silica particles on the surface. The removal of silica particles after the CMP process occurs according to the following reaction:



The other slurries that are mostly used for polishing glass substrates are CeO_2 or ZrO_2 . These slurries need stronger inorganic acids, such as nitric acid, in order to remove residue from the surface. It is also important to remember that a CMP process is only applied to a substrate before a lithography process. To truly understand the concept of cleaning, one needs to pay more attention to the mechanism of cleaning in more detail. In general, when discussing a cleaning process based upon dissolution, attention must be placed upon how that process actually occurs.

For example, when the HF molecule is utilized to remove silica particles, one of the by-products is SiF_6^{2-} . Such a by-product needs to be removed from the surface of particles immediately in order to allow fresh HF molecules to contact other layers of particles on the surface. Therefore, an agitation process may be used to facilitate this process. Otherwise, the byproduct of the HF and SiO_2 reaction (i.e., SiF_6^{2-}) will mask the outer surface of the remaining particles, and cleaning will not produce high yields.

Thus, in order to solve this problem, mechanical agitation systems are used, such as sonication, scrubbers, mechanical shakers, bobble agitators, and stream cleaning solutions instead of stationary solutions or high pressure spray tools. Another factor to keep in mind is keeping a balance between particle population and their sizes compared with the reactive concentration of species in the cleaning solutions. In general, there should always be more reactive molecules or ions in the cleaning solution, compared with the concentration of contamination.

To reach a high yield cleaning performance, it is necessary to understand the molecular structure of the surface and the chemistry of contamination before selecting a cleaning solution. It is always true that in the first step of selecting a cleaning solution, one will not find one cleaning solution that will meet all the cleaning requirements. However, by further pH optimization and/or components' concentration optimization in the cleaning solutions, it is possible to reach an acceptable process.

Another important factor in a wet cleaning process relates to the method that one chooses for the purpose of a cleaning process. Most fabrication sites currently use a stationary bath to clean their wafers or disks. Using a stationary bath for the cleaning process is economically quite sound, however, it has several

disadvantages as well. In a stationary bath, operators have less control over the cleaning process, because the active ingredients of the cleaning solutions are continuously consumed by the first layer of wafer or disk surfaces. Therefore, due to changes in concentration of the active species of the cleaning solution, it will be difficult for an operator to always keep the cleaning operation the same.

Another drawback of using a stationary bath is related to cross-contamination among good wafers or disks with contaminated parts. If, by accident, there are contaminated wafers or disks among other wafers or disks, the contaminated wafers or disks can contaminate the rest of the wafers and disks.

In the final step of the wet cleaning process, wafers or disks must also be rinsed by deionized (DI) water before being dried. However, the use of a stationary bath may affect the cleaning performance of wafers or disks at this step as well. When wafers or disks are processing in stationary cleaning bath, they carry a thin film of cleaning solution on their surfaces which may contain ionic contamination. When these wafers or disks are exposed to DI-water (immediate dilution of the cleaning solution), the wafers or disks will lose their surface protective shields that were created by the zeta potential from the cleaning solution. As a result, if there are any ion contaminants close to the surface, they can become bonded to the surface through electrostatic forces and create surface corrosion spots.

To solve this problem, there are three techniques that one may apply in order to eliminate rebonding of contamination to the surface during the DI-water rinsing process:

- Utilize a cleaning solution that contains a component with a low dissociation constant.
- Use a stream cleaning solution.
- Use a spray technique.

2.3.2.1 Cleaning Solution Containing a Component with a Low Dissociation Constant

With the first cleaning technique, where the cleaning solution can protect the surface from rebonding ions during DI-water rinsing, the cleaning solution must contain special components with low dissociation constants [3]. Such low dissociation molecules are mainly inactive in a high concentration cleaning solution because they cannot dissociate sufficiently to be active during the cleaning process. However, during the DI-water rinsing process (where the thin film of the cleaning solution on the wafers and disks is exposed to significant amounts of DI-water molecules), these inactive species within the cleaning solution become active and begin to build a protective shield on the surfaces of the wafers and disks during the first moments of the rinsing process. This protective shield, in the first critical moments of the rinsing process (the moment when

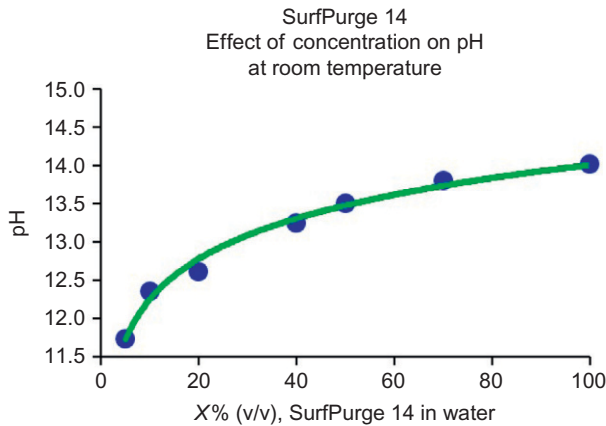


FIGURE 5.10 Effects of low dissociation compounds in cleaning solutions, such as SurfPurge 14, during the first moments of the DI-water rinsing process [3].

contaminant ions are attracted to the surface again), does not allow any ionic contaminants to approach the surface by forming a temporary zeta potential shield (Fig. 5.10).

2.3.2.2 Stream Cleaning Solution

The second technique is to have a stream cleaning solution flow inside the stationary bath. With this method, the flow of cleaning solution will carry the contaminants far away from the surface of the disks and wafers. Thus, after a certain cleaning period, when the wafers or disks are ready for DI-water rinsing, there is low risk of further rebonding of contamination. However, the cleaning cost with this method is higher than using a stationary bath. Therefore, in order to reduce the cleaning cost with this cleaning process, some fabrication sites have begun to recycle the cleaning solution through a filtration process. It is important to mention, if the operator decides to recycle the cleaning solution via filtration, the operator should be aware that regular filters are capable of trapping only certain sized particles that depend on the porosity size of the filter.

It should also be noted that filters cannot trap ions. If the by-products of a cleaning process are ionic species, then recycling of the cleaning solution through filters will not work because ions can easily pass through the filter pores. The only way to use a recycled cleaning solution is to use a combination of both regular filters for capturing particles and ion-exchange filters for capturing ionic species. However, using a combination of both standard filters to capture the particles and ion-exchange filters to trap ions from a stream flow bath has its own complexity.

In fact, this double filtration cannot be a very effective method if the concentration of some of the ions in the cleaning solution are significantly high.

Should this happen, the ion-exchange filter will be saturated and will not be effective. Thus, a combination of regular and ion-exchange filters usually works for only very low concentration of ionic species in the cleaning solution, such as removing certain low concentration ions from organic-based cleaning solutions, but not from inorganic-based cleaning solutions.

2.3.2.3 Spray Cleaning

The third method is to use a spray technique. With this method, the cleaning solution comes in contact with the surface of the wafers or disks by spray tools. Therefore, the surface is always in contact with a fresh solution and contaminants are washed away from the surface quite effectively. Although the usage cost of cleaning solutions with this technique is high, due to its cleaning concept and surface agitation, this method is quite effective.

2.4 Zeta Potential

The important rule for the zeta potential in removing ionic particles has been described. Yet, to have a better understanding of the cleaning process under the zeta potential, the interface of solid and liquid must be considered. Most electronic devices are fabricated on silicon (Si) substrate, with different thicknesses of silica (SiO_2) grown on the top surface in the form of a silanol group (Si-O-H). In most cases, hydrogen from the silanol group may be replaced with positive ionic charge particles, like SiOM.

Because the pH and concentration have significant roles in determining the value of the zeta potential [4], it is important to have a brief review on the interface of the solid (silica surface) and the cleaning solution. When the silanol group of the silica layer is in contact with positive particles or positive ions (cation), and if the attraction of these positive particles or positive ions towards the surface (SiO^-) is greater than the attraction of hydrogen ions with SiO^- , positive particles or positive ions can easily substitute for hydrogen ions of silanol groups in the form of SiOM (where M is a positive ion or a positive particle).

The first layer of ions that are in contact with the surface create a layer or plane that is called monolayer (Stern layer or Stern plane). The ions in this layer on the surface have low mobility toward the bulk solution, and their positions on the surface are fixed [5,6]. The second layer of ions that are attracted to the top of the monolayer create another layer or plane that is called the slipping plane (SP). The combination of mono and the second layer is called the double layer. Ions in the second layer are more mobile compared with ions in the monolayer. The zeta potential is the potential between the SP and the surface of the solid.

Knowledge about the zeta potential brings detailed insight into the causes of dispersion, aggregation, or flocculation, and may be applied to improve the formulation of dispersions, emulsions, and suspensions of particles of interest.

One of the physical properties of particles in the state of suspension is the zeta potential. This property may be used to understand the degree and stability of suspensions and emulsions. A colloidal system is formed by dispersion of one of the physical states of matter: gas, liquid, and solid in a continuous phase of a different state. Characteristics of colloidal suspension are very important in modern technology. Any adhesion forces between particles, form aggregates, and sometimes coagulation phases; it may settle under the force of gravity.

Based on the DLVO Theory (Derjaguin & Landau and Verwey & Overbeek [7]), the stability of a colloidal system depends on its total potential energy (V_t), defined as:

$$V_t = V_a + V_r + V_s \quad (1)$$

V_a and V_r are van der Waals attractive and repulsive forces between particles and their environments, respectively. V_s stands for the potential energy of a solvent. The attractive force " V_a " can be defined as:

$$V_a = -A/(12\pi D^2) \quad (2)$$

Here, " A " is the Hamaker constant and " D " is the particle separation for distance. The repulsive force " V_r " is also related to the zeta potential as seen by the following equation:

$$V_r = 2\pi\epsilon R\zeta^2 \exp(-\kappa D) \quad (3)$$

Here, " R " is the particle radius, " ϵ " is the dielectric constant, " κ " is a parameter which is a function for the ionic composition, and " ζ " is the zeta potential.

Due to the Brownian or thermal motion, when particles with similar charges approach each other, V_r prevents the two particles from adhering together. It means that the balance between V_a and V_r stabilizes the colloidal system. However, if the particles have enough energy to overcome the energy barrier, the attractive force will pull them into close contact where they will adhere together. Based on this statement, the stability of a colloidal system depends upon the repulsive force between the particles.

To stabilize a colloidal system, the van der Waals force must be reduced by adding surfactants to the colloidal dispersion. Surfactants will coat the particles and increase the steric repulsion between them, and thus reduce the van der Waals force. The second approach to stabilize the colloidal particles is by the electrostatic stabilization method. From the brief introduction above, the zeta potential is considered an index of the magnitude of the interaction between the colloidal particles. By knowing the zeta potential of the solution, one will be able to predict the stability of the system.

The charges on the surface of the ionic particles within the cleaning solutions depends on the concentration of hydronium ions (H_3O^+) or the hydroxide ions (OH^-), close to the particles. If the particles carry an acidic functional group, the dissociation of the acidic group will generate negatively charged

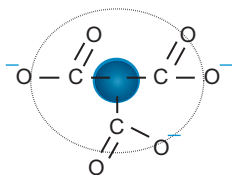


FIGURE 5.11 Result of dissociation of the acidic functional group of particle.

particles (Fig. 5.11), and the dissociation of the basic groups will generate positively charged particles. This means that the pH of the solution affects the magnitude of the particle surface charge density. Therefore, the surface charge of the particles may be adjusted by varying the pH of the solution.

Adding an ionic surfactant may change the particle charge by desorption. The development of a net charge on the particle surface by ionic surfactant will change the charge distribution around the particle, and as a result, create an electrical double layer around each particle.

The double layer around each particle consists of inner (Stern) and outer (diffuse) layers. The double layer will move with the displacement of the particles. And the potential at this boundary (hydrodynamic shear) is called the zeta potential, where the microscopic shear force is caused by the motion of different layers in a fluid.

In a suspension system, if all particles have a large negative or positive zeta potential, they will repel each other (Fig. 5.12). However, a low zeta potential causes the particles to come together and the suspension is flocculated.

In general, the zeta potential between a stable and an unstable suspension system may be ± 30 mV. Particles with the zeta potential that are more positive than +30 mV or more negative than -30 mV, are considered stable.

2.4.1 Factors Affecting the Zeta Potential

2.4.1.1 pH

Adding acid or base builds up positive or negative charges on the particles which can change the zeta potential. If the zeta potential approaches zero, this point is called the isoelectric point, where the colloidal system is least stable.

2.4.1.2 Effect of Concentration

The thickness of the double layer (κ^{-1}) around the particle depends on the concentration of ions in the solution and can be calculated from the ionic strength I of the medium as

$$I = \frac{1}{2} \sum_{i=1}^n c_i z_i^2 \quad (4)$$

Here, “ I ” is ionic strength, “ c_i ” is concentration, and “ z_i ” is ionic charge.

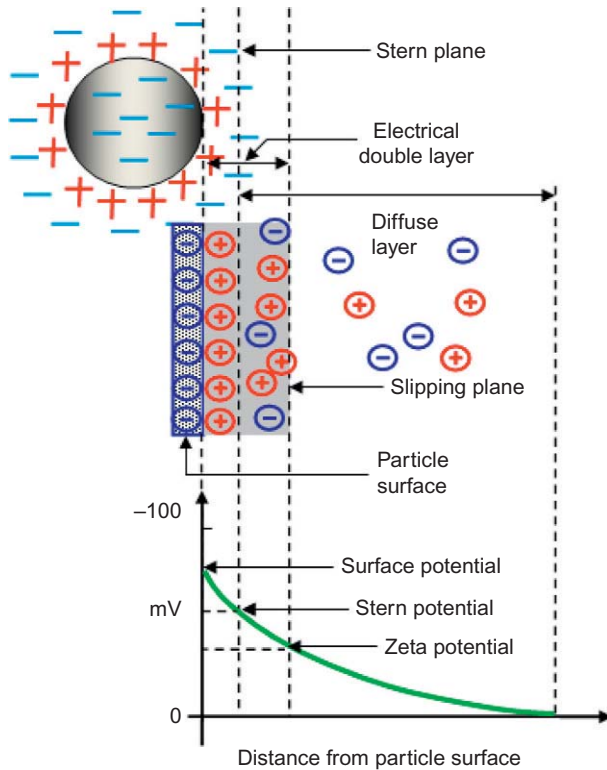


FIGURE 5.12 Representation of double layer, diffusion layers, and the zeta potential.

It is obvious that the valence of the ions (z_i) will influence double layer thickness. A trivalent ion, such as Fe^{3+} , will compress the double layer to a greater extent in comparison with a monovalent ion such as K^+ . If any type of ion adsorbs on the surface of the particles, it can change the zeta potential dramatically.

2.4.2 Measurement of the Zeta Potential

The zeta potential cannot be measured directly. Determination of the zeta potential is related to the measurement of mobility of the particles by methods such as electrophoretic mobility and using theoretical models. The motion of dispersed particles relative to a liquid under the influence of a spatially uniform electric field is called electrophoresis.

In general, systems that are used for measuring the zeta potential illuminate particles with laser lights. As a result, the illuminated particles will begin to scatter the light. The frequency of scattered light by illuminated particles

is a function of particle velocity (Doppler shift) [8]. A reference beam is used to determine the frequency shift in the scattered light by particles. By measuring the particle velocity (cm/s), one can determine the mobility of the particles as $\text{cm}^2/\text{v.s}$, where “v” is the applied voltage and “s” is the time (seconds).

The most common model, is called the Smoluchowski Model [9], can calculate the zeta potential if other physical factors, such as liquid dielectric constant, refractive index, and viscosity of liquid, are known.

3 CLEANING OF ORGANIC CONTAMINANTS

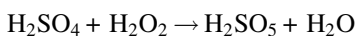
The main source of organic contamination, other than fingerprints, is the application of photoresist as a masking process, traces of wax after CMP, or application of certain adhesives during the packaging process. Fingerprints are present most often when gloves are dirty, or wafers are handled by bare hands. However, fingerprint contamination can easily be removed by immersing the wafers or disks in hot sulfuric acid (provided no metallization lines are present on the wafers) or by immersing them in organic solvents such as isopropyl alcohol.

Photoresists are used in electronic device fabrication as a temporary mask during the lithography process. In general, regardless of their chemical structures, photoresists may be divided into two different groups: positive and negative photoresists. In a positive photoresist, the sections of the photoresist that are exposed to the UV light become soft, while the nonexposed sections remain hard and will protect the underneath layers of an electronic device. In a negative photoresist, this situation is reversed.

Cleaning traces of photoresists is a bit more complex. They may be removed by using any of the following processes:

- Photoresist stripping by wet inorganic process.
- Photoresist stripping by wet organic process.
- Photoresist stripping by dry process.

In general, in the absence of any metal lines on an electronic device, the whole or traces of positive photoresist can be cleaned with the wet cleaning process using a mixture of hot concentrated sulfuric acid and hydrogen peroxide. In this process, the reaction of high concentration of sulfuric acid, with hydrogen peroxide at a high temperature, creates a strong oxidizing agent (H_2SO_5) that is able to oxidize and dissolve the photoresist very easily.



This process may also be carried out by exposing the photoresist to ozone gas bubbling in concentrated sulfuric acid. This photoresist stripping process is considered to be the cheapest way to strip photoresist in the absence of any electronic metal lines on the device. If the electronic device is in the critical stage of

its lithography process with metallization lines, the photoresist may also be stripped by using organic solvents or solutions.

The most common photoresist strippers are made of amine solutions or solvents, such as tetra-methyl ammonium hydroxide (TMAH) or *N*-methyl 2-pyrrolidone. There are also acidic photoresist strippers with the main component being methyl sulfonic acid or its derivatives. In some cases, when the photoresist is not baked in high temperatures, a simple organic solvent such as acetone may also be used to dissolve and remove the photoresist mask. Usually, most organic photoresist strippers contain corrosive inhibitors that can protect the metallization section of electronic devices.

By using an organic solvent or organic solution, the cleaning mechanism cannot be carried out very precisely. This is because all wet cleaning processes cause some level of undercutting. When an organic photoresist stripper is used in the presence of metals, metal lifting can occur, depending on the photoresist baking temperature, photoresist thickness, and also the way the metal layers, such as aluminum, copper, or gold, are fabricated.

When an electronic device becomes very dense, operators use another cleaning process called the ASH process (using plasma sources). In an ASH process, the photoresist is exposed to reactive ions that are capable of reacting with the photoresist, and breaking down its structure into different gases. With this technique, the process time and temperature and type of reactive ions must be controlled before the reactive ions reach the surface of the metal ions. Although the ASH process is one of the most common practices for photoresist stripping, sometimes these reactive ions will create some hard-to-clean polymer molecules. Hard polymers will form after the ASH process when reactive ions react with the photoresist in the presence of silica. These hard polymers may be removed by solutions containing low concentrations of HF in an appropriate organic solvent (Fig. 5.13).

4 EFFECTS OF SURFACTANTS IN CLEANING SOLUTIONS

Surfactants have a special role in the cleaning process by lowering the surface tension of the solution. In general, surfactant activity mostly affects the interaction of the cleaning solution with the substrate surface. Most surfactants are organic compounds with a hydrophobic group on one side (tail), and a hydrophilic group on the other side (head), as shown in Fig. 5.14. The coexistence of two different functional groups (ionic and nonionic) in this type of molecular structure makes these molecules soluble in both water (a polar liquid), as well as in organic solvents with a lower polarity characteristic compared to water. Adding surfactant to two nonmiscible liquids (such as water and oil) would produce an emulsion. Therefore, cleaning solutions must be formulated in such a way that the mixture of all of the chemical compositions in a cleaning solution are homogeneous.

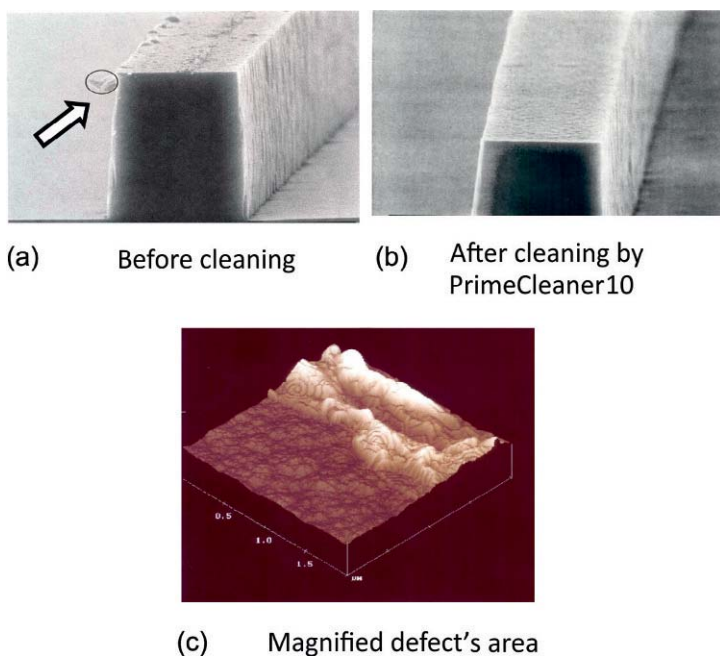


FIGURE 5.13 Effects of Solexir PrimeCleaner10 on cleaning hard to remove polymers after dry etch [10]. (a) Before the defect is exposed to cleaning solution. (b) After the defect is exposed to cleaning solution. (c) Magnified defect on the surface.

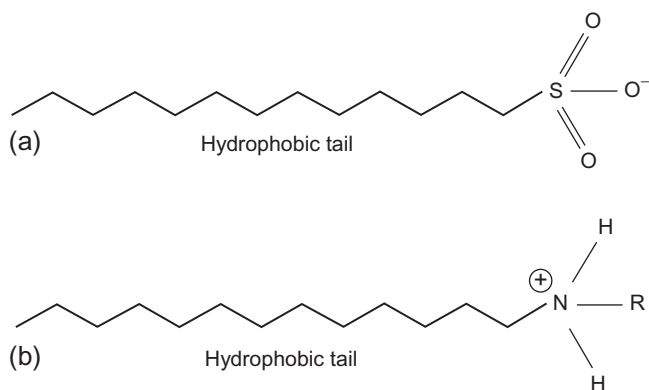


FIGURE 5.14 Surfactant structure (a) with negatively charged head and (b) with positively charged head.

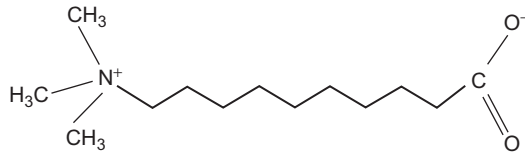


FIGURE 5.15 Example of zwitterion surfactant containing both positive and negative heads.

Surfactants themselves may be divided into two major groups: ionic surfactants and nonionic surfactants. An ionic surfactant is a surfactant that carries an ionic group in the form of either a negative [11–13] or a positive head [14,15]; or it may carry both simultaneously in one molecule (zwitterions) (Fig. 5.15).

Depending upon the ionic charge of the surfactant functional group and its concentration within the solution, the ionic head of the surfactant will be attracted to the opposite charge. In principle, when an ionic charge of a surfactant is attracted to the functional group of the surface, it causes charge density reduction in the surface functional group. As a result, the attraction force between the electrostatically bonded contamination particles with the surface functional group decreases and these particles may be removed more easily. Therefore, the surface tension of each surfactant will vary depending on its molecular structure and the chemical composition of the cleaning solution.

5 WET CLEANING FOR REMOVING TRACE METALS AFTER ETCHING PROCESS

During the fabrication process, cleaning active and inactive sections of each electronic device is very important. Therefore, the concept of a clean surface is applied to a surface free of contamination. Within any cleaning process, removing traces of metals after an etch process is as important as cleaning particles or traces of photoresist, or traces of slurries, after a CMP process. Any leftover traces of metal may easily cause a short circuit, as well as corrosion, both of which may make the electronic device nonfunctional.

In the etching process, the surface that comes in contact with the etchant chemical is removed (see Fig. 5.16). Often, an etching process is either

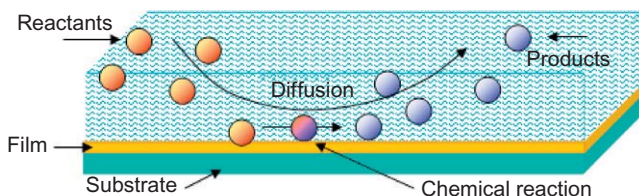


FIGURE 5.16 Diffusion of etchant ions toward the substrate surface and dissolution of trace metal left over after the etching process.

diffusion-rate limited or reaction-rate limited [16]. In the diffusion-rate limited etching process, the etch rate depends on how fast reactants can reach the surface, as well as how fast the by-products of an etching process are removed. In contrast, in a reaction-rate limited process, the rate of reaction is only limited to the rate of chemical reactions of the etchant with the surface.

It is clear that other physical factors, such as temperature, agitation, and concentration of the etchant species in the etchant solution, can significantly affect the etch rate, as well as the uniformity of the top surface layer.

Wet chemical etching can occur either simply by dissolution, or it can occur in two or more steps, involving oxidation, followed by dissolution (such as the aluminum etching process). In this process, first, the etchant will diffuse to the surface of the substrate, and, second, it reacts with the surface by any means, such as oxidation, and then dissolves the unprotected area of the substrate where the mask or photoresist is not covered; and finally, the by-products of etching are removed from the surface by the agitation process in order to create an open space for the fresh etchant molecules [17].

Formation of gas bubbles on the metal surface, as a result of the reaction between the acid in the etch solution and the metal, is an important issue during the etching process. If these gas bubbles are not removed from the surface, they can result in non-etching, or in incomplete etching.

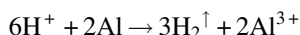
In the following sections, a brief cleaning related to the most important components that are used in fabrication of electronic devices are explained. Therefore, it is important to pay attention to chemical composition, chemical concentration, as well as process temperature for each solution.

5.1 Aluminum

Wet etching of aluminum is often done in a heated solution of phosphoric acid, acetic acid, nitric acid, and water. A typical composition may be 80% phosphoric acid, 5% acetic acid, 5% nitric acid, and 10% water—typically operated at a temperature of 45–50 °C. An etch rate of ~300–500 nm/min may be achieved by controlling the etchant composition, process temperature, agitation, and the impurities or alloys in a predominantly Al film.

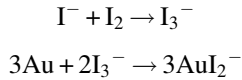
In the aluminum etching process, nitric acid converts Al to Al₂O₃ with the evolution of hydrogen gas. The aluminum oxide will then dissolve in a phosphoric acid and the rest of the nitric acid, in order to produce aluminum phosphate and aluminum nitrate. The hydrogen bubbles formed by this reaction tend to stick to the Al surface, leading to an etch nonuniformity. Agitation and an optional surfactant in the etch solutions will then help release the gas bubbles from the aluminum surface, and as a result, acids will etch aluminum thin films smoothly [18,19].

The overall etch reaction is written as follows:



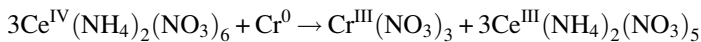
5.2 Gold

The etching of Au may be done with aqua regia (HCl:HNO₃, with a volume ratio of 3:1), potassium cyanide (KCN), and potassium iodine complex (KI₃) [20]. The iodine solution (I₂/KI or KI₃) is often viewed as the most stable and safest method of etching gold [21]. The etching reactions may be written as follows:



5.3 Chromium

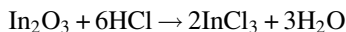
The most common Cr wet etchant is based on the aqueous solution of cerium ammonium nitrate (CAN) chemistry, with added nitric or perchloric acid as an oxidizer. CAN oxidizes Cr to Cr³⁺, while cerium is reduced from Ce (IV) to Ce (III):



Cr etching is often done at room temperature [22], and would typically take 1 minute or less to etch a Cr film at a thickness of 100 nm.

5.4 Indium Tin Oxide

Indium tin oxide (ITO) is often etched with hydrochloric acid solutions, or premixed with nitric acid. Depending upon the volume ratio of amorphous ITO, versus crystalline phases of ITO, the etch rate, etch uniformity, and the extent of post-etch residues will vary [23].

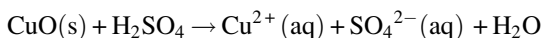
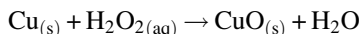
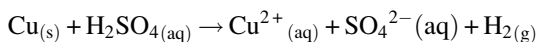


5.5 Titanium Tungsten

Titanium tungsten (TiW) is often etched using 30% hydrogen peroxide. The etch rate at room temperature is slow—approximately 10 nm/min. The etch rate may be increased to ~40 nm/min by operating the peroxide at 50 °C. Using a mixture of ammonium hydroxide and hydrogen peroxide would be the fastest method of etching TiW, but the peroxide is destabilized in an alkaline pH environment [20,24].

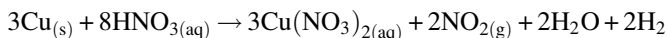
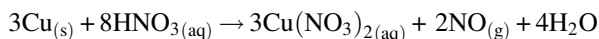
5.6 Copper

Besides the ferric chloride and cuprous chloride type etchants, Cu is most often etched by using concentrated sulfuric acid or a mixture of diluted sulfuric acid and hydrogen peroxide as follows [25–28]:



Copper can also dissolve in a mixture of acetic acid and hydrogen peroxide, or in a mixture of acetic acid, phosphoric acid, and hydrogen peroxide [29].

Diluted and concentrated nitric acid based etchants have also been utilized for copper etching according to the following reactions:



Depending on the etch selectivity, film thickness and other requirements, an appropriate etching product should be selected.

5.7 Nickel

Nickel is most often etched with a 15-20% aqueous solution of nitric acid, among a few other possible etchants [23]. If a Ni oxide film is present on the surface, it is often removed by using a 10% aqueous solution of hydrochloric acid. For Ni and its oxide, the addition of an appropriate surfactant in the etchants mixture will also help with a more effective and faster removal of the material from the surface [30].

5.8 Gallium Arsenide

There are many etchants for gallium arsenide (GaAs); however, very few can be used for the isotropic etching process. Usually, As is more reactive with the etching solution compared to Ga, and, as a result, it can etch faster than Ga. The difference between the etch rate of As versus Ga in an etching solution, is that Ga usually leaves a defect on the surface of GaAs.

Almost any combination of an oxidizer (such as hydrogen peroxide) and oxide dissolver (such as acid or base) may be used as an etchant for GaAs. For example, mixtures of sulfuric acid and hydrogen peroxide, or ammonium hydroxide and hydrogen peroxide, are often used for etching GaAs. Another common GaAs etching solution may be phosphoric acid, hydrochloric acid, or citric acid mixed with hydrogen peroxide [31].

6 SILICON PRODUCT WET ETCHING PROCESS

In the following sections, the etching of silicon/silica products in the form of isotropic and the anisotropic process, or cleaning of residue from silicon/silica products, as well as other thin films of silicon compounds such as silicon nitride,

are briefly explained. In addition, the etch selectivity as well as etch rate for use of different etch/clean solutions are also discussed.

6.1 Silica Isotropic and Anisotropic Etching

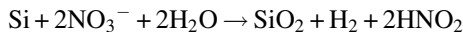
The wet etching process is either isotropic (orientation independent) or anisotropic (orientation dependent), as shown in Fig. 5.17. Usually, most wet etching processes are isotropic, which are adequate for geometries of greater than 3 μm . In isotropic wet etching [32], material is removed uniformly from all directions by HF or buffered HF solutions (NH_4F and HF mixture). On the other hand, in anisotropic etching techniques, material is removed uniformly from a vertical direction (more control within vertical walls, fewer masks undercut). Details of the anisotropic etching process can be found in Section 6.4.

Wet etching is often perceived as low cost, a reliable method and is suited for high production environments with high selectivity in many cases. For a completely anisotropic etch, where small geometry devices are the etch candidates, the preference is to use dry etching, which creates straight vertical walls.

For a wet etch, it is important that one selects a mask that does not dissolve in the etching solution, or at least the rate of its dissolution in the etchant is slower than the layers targeted to be etched.

6.2 Silicon Isotropic and Anisotropic Etching Process

Silicon is etched by a nitric and hydrofluoric acid mixture (HNO_3 may be replaced by other strong oxidants like H_2O_2). In this process, HNO_3 partially decomposes to HNO_2 , which can oxidize the surface of Si to SiO_2 :



After formation of silica, the HF reacts with SiO_2 to form H_2SiF_6 by the following overall reaction:

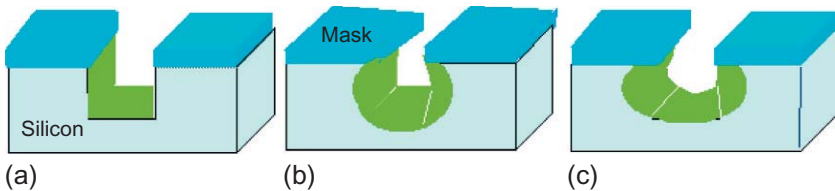
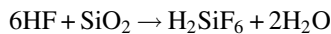


FIGURE 5.17 Comparison of dry etching and wet etching processes for silica. (a) Complete anisotropic etching process with straight vertical walls without mask undercut. (b) Partial anisotropic etching process with minor mask undercut. (c) Complete isotropic etching process with excess mask undercut.

The excess nitric acid results in excessive SiO_2 formation, and the etch rate becomes limited by the ability of HF to remove the oxide (polishing process).

From the chemical reaction above, one can see hydrogen gas as the by-product from the first reaction, which is formed on the surface of the silicon as bubbles. Therefore, it is very important to immediately remove hydrogen bubbles from the surface by an agitation process. Allowing the hydrogen gas to remain on the silicon surface during the etching process prevents fresh HF molecules to react with the silicon covered with hydrogen bubbles, thus, creating a nonuniformly etched metal surface.

Two alkaline solutions, potassium hydroxide (KOH) and tetramethylammonium hydroxide (TMAH), are normally used as the etchants for the anisotropic wet process. KOH has excellent uniformity and reproducibility characteristics [33], but it is not compatible with electrical circuits due to the high diffusion rate of potassium ions in the silicon or silica, and its chemical reactions with metallization sections of electronic devices because of its high pH.

The silicon etch rate by KOH strongly depends upon the crystallographic orientations of the silicon. This is why the production of the 3D microstructures is possible by utilizing KOH as an anisotropic wet etching chemical [34]. The overall chemical etching reaction by the alkaline solution is as follows:



Such a reaction would precede at a lower rate in the absence of any one of the two reactants in the equation above, namely, water molecules (H_2O) or hydroxide ions (OH^-).

This is because at a low KOH concentration, OH^- is in low concentration, and at a high KOH concentration, H_2O is in low concentration; therefore, the reaction would slow down at both high and low KOH concentrations. However, this is by no means a comprehensive explanation of all the experimental results obtained.

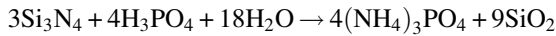
TMAH is an organic basic solution that may be used for anisotropic etching of silicon [35,36], with a typical etch rate of 0.5-1.5 $\mu\text{m}/\text{min}$. Due to a lack of high diffusing ions, such as sodium and potassium ions, the use of this solution for silicon etching is preferred. The process temperature for TMAH, similar to potassium hydroxide, is high, about 80-90 $^\circ\text{C}$. However, the selectivity of TMAH over KOH for different silicon orientations is much lower. For example, the selectivity of TMAH for silicon orientation $\langle 111 \rangle / \langle 100 \rangle$ is approximately 10-35 times lower than KOH.

Looking at other aspects of TMAH and KOH, it is possible to see that both can attack aluminum lines, and cannot be used for etching silicon in the presence of aluminum. Also, both TMAH and KOH may be used for the boron-stop etching process. Results show that the etch rate of silicon with boron doped in concentration of $10^{20}/\text{cm}^3$ is approximately 40 times lower for TMAH. In the lithography process, TMAH is mostly used as a positive photoresist stripper due to its high pH and lack of diffuser ions.

In general, both isotropic and anisotropic wet etching demonstrate relatively high etch rates for an isotropic etch compared with an anisotropic etch [37]. Most applications of an anisotropic etch are in manufacturing of solar cells, anti-reflecting surfaces, and fabrication of waveguides, infrared detectors and high value capacitors [38].

6.3 Silicon Nitride

Si_3N_4 is often etched with a reflux boiling 85% phosphoric acid at 180 °C. Under such conditions, photoresist may be lifted because Si_3N_4 often utilizes a thin layer of oxide in order to mask the nitride [39,40]. The overall reaction may be depicted as follows:



In general, Si_3N_4 can also be etched very slowly by HF solutions at room temperature.

6.4 Dry Etching Process

As mentioned before, dry etching is usually used for small geometry devices in order to have more control in shaping the vertical walls. Most reactive ions that are created from chemical dry etching are tetrafluoromethane (CF_4), sulfur hexafluoride (SF_6), chlorine gas (Cl_2), or fluorine gas (F_2) [41]. In the dry etching process, control of process time, process temperature as well as selection of the reactive ion mixture, are very important. In most cases, dry etching is followed by a wet cleaning process to remove the residue of dry etching by-products.

During the dry etching process, when using a reactive ion etchant (RIE), the vacuum chamber must continuously be furnished with the selected etchant gases. The radio frequency (RF) power creates plasma which is a mixture of radicals and ions. Reactions of these active high energy species with the surface of wafers create some by-products in the form of gases or other volatile species that will subsequently desorb from the wafer surface and evacuated through the vacuum exhaust, allowing fresh reactive ion etchants into the vacuum chamber. Most typical dry etchants for Si, as mentioned previously, are CF_4 , SF_6 , and $\text{BCl}_2 + \text{Cl}_2$ with an etch rate of ~ 50 nm/min. For SiO_2 , the etchants are a mixture of $\text{CHF}_3 + \text{O}_2$ or $\text{CF}_4 + \text{H}_2$, with an etch rate of 20 nm/min. For Si_3N_4 , the etchant is a mixture of CF_4 , with O_2 or H_2 or CHF_3 and an etch rate of 10 nm/min.

Usually during a dry etching process, when the pressure inside the vacuum chamber decreases to less than 10 mm Hg, the beam energy will be very high and etch performance will be very good for anisotropic orientations (straight vertical wall). But this condition creates poor selectivity.

If the pressure of the vacuum chamber reaches higher than 100 mmHg, the beam energy and anisotropic performance becomes very low (not a completely straight vertical wall), but the selectivity of this process will be excellent.

6.5 Advantages and Disadvantages of Dry Versus Wet Etching

In general, the difference between wet and dry etching may be summarized as follows: all wet etching can be conducted in normal atmospheric environment, whereas dry etching can be performed only in a vacuum. Low operation cost, high etching rate, and good selectivity for most materials are advantages of the wet process for larger device geometries; whereas, the fabricating capability of electronic devices with features less than 100 nm, is an advantage of the dry etching process.

The disadvantage of the wet etching process is related to inadequate power of wet etching to fabricate an electronic device with a feature size $< 1\ \mu\text{m}$, creating significant amounts of hazardous waste, and the risk of wafer contamination. The disadvantages of the dry etching process are related to the high cost of the etching process, its poor selectivity, and the potential of surface damage due to exposure of wafers to RF radiation in the vacuum chamber.

From the discussion in the previous sections, understanding the chemical nature of contaminants and selection of the right cleaning process with respect to the desired geometry is very important. As has been explained above in the advantages and disadvantages of different cleaning procedures, sometimes selection of a cleaning process with low cost is not a good idea because it can create a finished product with higher cost due to a low production yield.

With regard to the etching process, one should remember to use the same etching chemical compositions for cleaning the leftover residue of contaminants after the etching process. One can use lower concentrations of the etching chemicals to get rid of these residues, but in general, the best results will be obtained if one uses the same chemical composition.

7 SUMMARY AND CONCLUSIONS

The most important factors facing a semiconductor process engineer is when he/she attempts to clean a surface, following lithography steps, which can be summarized as follows: first, one must understand, and has to pay attention to the molecular structure of the surface that requires cleaning. Knowing the type of contamination, nature of the contamination (organic or inorganic), and its chemical bonding to the surface is crucial in developing a successful cleaning process strategy.

Secondly, knowledge of chemical cleaning solutions for removing surface contaminants is very important. It is extremely important to know in advance whether or not new sources of contamination (by-products of cleaning) could occur through either chemical interaction of the cleaning solution with the surface or via a dissolution process.

Although the right selection of a cleaning solution is very important, this choice may be dramatically affected if the engineer in charge of cleaning the surface uses the wrong cleaning equipment or process; for example, by applying

a stationary bath versus a stream bath or spray tools. Therefore, in this chapter, different types of contamination such as ionic and nonionic, along with possible cleaning methods with their advantages and disadvantages, are discussed. In addition, the cause of rebounding contamination during water rinsing, where the substrate loses its zeta potential protecting shield, is discussed.

Also, in this chapter, we address the importance of zeta potential and its dependency on pH, and the effects of surfactants, inhibitors, and chelating agents. Finally, a brief review of cleaning solutions for removing traces of metal residue and photoresist is also provided.

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